ANSWER KEY & MARKING SCHEME SET 1

Chapter 9: **Symmetry** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 2	Along its two diagonals.
2	**(d)** 72	$360^\circ \div$ number of sides.
3	**(d)** the same letter	Vertical-axis symmetry keeps it the same.
4	**(d)** 4	Order = $360 \div$ smallest angle.
5	**(c)** 7	Sides = lines of symmetry.
6	**(a)** Z	Z fits after a half turn but cannot be folded.
7	**(b)** 8	Order $= 360^\circ \div$ angle.
8	**(a)** 0	Symmetry property
9	**(a)** 60	$360^\circ \div$ number of sides.
10	**(b)** the same letter	Vertical-axis symmetry keeps it the same.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Fold the figure so that the two halves coincide; There are 2 lines of symmetry (1); Lines pass through vertices/midpoints as per the shape (1).	1+1
12	Angle $= 360 \div 5$ (1); 72° and order 5 (1).	1+1
13	Reverse the order of letters and reflect each letter; Reversed order: MAM (1); Looks the same (1).	1+1
14	Fold the figure so that the two halves coincide; There are 2 lines of symmetry (1); Lines pass through vertices/midpoints as per the shape (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sides $= 360 \div 40 = 9$ (1); Lines of symmetry $= 9$ (1); Order of rotational symmetry equals the number of sides (1).	1+1+1
16	(i) Vertical line: yes (1); (ii) Horizontal line: no (1); (iii) Rotational symmetry of order 2: no (1).	1+1+1
17	Image is the same distance behind the mirror (1); Image is 3 units below the line (1); Distance $= 6$ units (1).	1+1+1
	OR: Sides $= 360 \div 45 = 8$ (1); Lines of symmetry $= 8$ (1); Order of rotational symmetry equals the number of sides (1).	1+1+1
18	(i) Vertical line: yes (1); (ii) Horizontal line: no (1); (iii) Rotational symmetry of order 2: no (1).	1+1+1
	OR: Image is the same distance behind the mirror (1); Image is 4 units below the line (1); Distance $= 8$ units (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(a) 6 (1); (b) 6 (1); (c) 60° (1); (d) $4 \times 60 = 240^\circ$ (2).	1+1+1+2
	OR: Letters with vertical symmetry: M, A, T, H (1); Letters with horizontal symmetry: H (1); Order-2 rotational letters: H (1); Word symmetric (vertical): no (2).	1+1+1+2
20	(a) 8 (1); (b) 8 (1); (c) 45° (1); (d) $3 \times 45 = 135^\circ$ (2).	1+1+1+2
	OR: Letters with vertical symmetry: T, O, O, T (1); Letters with horizontal symmetry: O, O (1); Order-2 rotational letters: O, O (1); Word symmetric (vertical): yes (2).	1+1+1+2

Mistakes to avoid:

- Saying a rectangle has 4 lines of symmetry (diagonals are not lines of symmetry); a non-square rectangle has only 2.
- Giving a parallelogram 2 lines of symmetry; it has none, though it has rotational symmetry of order 2.
- Mixing up the order of rotation with the smallest angle: angle $= 360^\circ \div$ order.
- Saying a figure with order 1 has rotational symmetry; order 1 means only the trivial full turn.
- Drawing the mirror image at the wrong distance from the mirror line or on the same side of it.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 9: **Symmetry** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 5	Order = $360 \div$ smallest angle.
2	**(d)** 9	Sides = lines of symmetry.
3	**(d)** H	H has two fold lines.
4	**(c)** 6	Order $= 360^\circ \div$ angle.
5	**(c)** 1	Symmetry property
6	**(c)** 40	$360^\circ \div$ number of sides.
7	**(a)** the same letter	Vertical-axis symmetry keeps it the same.
8	**(d)** 6	Order = $360 \div$ smallest angle.
9	**(d)** 11	Sides = lines of symmetry.
10	**(d)** C	Fold along the horizontal middle.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Angle $= 360 \div 9$ (1); 40° and order 9 (1).	1+1
12	Reverse the order of letters and reflect each letter; Reversed order: TOT (1); Looks the same (1).	1+1
13	Fold the figure so that the two halves coincide; There are 1 lines of symmetry (1); Lines pass through vertices/midpoints as per the shape (1).	1+1
14	Angle $= 360 \div 10$ (1); 36° and order 10 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sides $= 360 \div 60 = 6$ (1); Lines of symmetry $= 6$ (1); Order of rotational symmetry equals the number of sides (1).	1+1+1
16	(i) Vertical line: no (1); (ii) Horizontal line: no (1); (iii) Rotational symmetry of order 2: yes (1).	1+1+1
17	Image is the same distance behind the mirror (1); Image is 5 units below the line (1); Distance $= 10$ units (1).	1+1+1
	OR: Sides $= 360 \div 72 = 5$ (1); Lines of symmetry $= 5$ (1); Order of rotational symmetry equals the number of sides (1).	1+1+1
18	(i) Vertical line: no (1); (ii) Horizontal line: no (1); (iii) Rotational symmetry of order 2: yes (1).	1+1+1
	OR: Image is the same distance behind the mirror (1); Image is 6 units below the line (1); Distance $= 12$ units (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(a) 5 (1); (b) 5 (1); (c) 72° (1); (d) $2 \times 72 = 144^\circ$ (2).	1+1+1+2
	OR: Letters with vertical symmetry: W, O, W (1); Letters with horizontal symmetry: O (1); Order-2 rotational letters: O (1); Word symmetric (vertical): yes (2).	1+1+1+2
20	(a) 9 (1); (b) 9 (1); (c) 40° (1); (d) $4 \times 40 = 160^\circ$ (2).	1+1+1+2
	OR: Letters with vertical symmetry: O, O (1); Letters with horizontal symmetry: O, O (1); Order-2 rotational letters: N, O, O, N (1); Word symmetric (vertical): no (2).	1+1+1+2

Mistakes to avoid:

- Saying a rectangle has 4 lines of symmetry (diagonals are not lines of symmetry); a non-square rectangle has only 2.
- Giving a parallelogram 2 lines of symmetry; it has none, though it has rotational symmetry of order 2.
- Mixing up the order of rotation with the smallest angle: angle $= 360^\circ \div$ order.
- Saying a figure with order 1 has rotational symmetry; order 1 means only the trivial full turn.
- Drawing the mirror image at the wrong distance from the mirror line or on the same side of it.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 9: **Symmetry** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 5	Order $= 360° \div$ angle.
2	**(d)** 1	Symmetry property
3	**(a)** 36	$360° \div$ number of sides.
4	**(c)** the same letter	Vertical-axis symmetry keeps it the same.
5	**(c)** 8	Order = $360 \div$ smallest angle.
6	**(a)** 12	Sides = lines of symmetry.
7	**(b)** F	No line, no rotation.
8	**(b)** 4	Order $= 360° \div$ angle.
9	**(b)** 2	Symmetry property
10	**(b)** 45	$360° \div$ number of sides.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Reverse the order of letters and reflect each letter; Reversed order: WOW (1); Looks the same (1).	1+1
12	Fold the figure so that the two halves coincide; There are 1 lines of symmetry (1); Lines pass through vertices/midpoints as per the shape (1).	1+1
13	Angle $= 360 \div 12$ (1); $30°$ and order 12 (1).	1+1
14	Reverse the order of letters and reflect each letter; Reversed order: TAH (1); Does not look the same (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sides $= 360 \div 90 = 4$ (1); Lines of symmetry $= 4$ (1); Order of rotational symmetry equals the number of sides (1).	1+1+1
16	(i) Vertical line: yes (1); (ii) Horizontal line: no (1); (iii) Rotational symmetry of order 2: no (1).	1+1+1
17	Image is the same distance behind the mirror (1); Image is 7 units below the line (1); Distance $= 14$ units (1).	1+1+1
	OR: Sides $= 360 \div 120 = 3$ (1); Lines of symmetry $= 3$ (1); Order of rotational symmetry equals the number of sides (1).	1+1+1
18	(i) Vertical line: no (1); (ii) Horizontal line: no (1); (iii) Rotational symmetry of order 2: yes (1).	1+1+1
	OR: Image is the same distance behind the mirror (1); Image is 8 units below the line (1); Distance $= 16$ units (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(a) 10 (1); (b) 10 (1); (c) $36°$ (1); (d) $3 \times 36 = 108°$ (2).	1+1+1+2
	OR: Letters with vertical symmetry: O, X, O (1); Letters with horizontal symmetry: O, X, O (1); Order-2 rotational letters: O, X, O (1); Word symmetric (vertical): yes (2).	1+1+1+2
20	(a) 12 (1); (b) 12 (1); (c) $30°$ (1); (d) $5 \times 30 = 150°$ (2).	1+1+1+2
	OR: Letters with vertical symmetry: H, I, M (1); Letters with horizontal symmetry: H, I (1); Order-2 rotational letters: H, I (1); Word symmetric (vertical): no (2).	1+1+1+2

Mistakes to avoid:

- Saying a rectangle has 4 lines of symmetry (diagonals are not lines of symmetry); a non-square rectangle has only 2.
- Giving a parallelogram 2 lines of symmetry; it has none, though it has rotational symmetry of order 2.
- Mixing up the order of rotation with the smallest angle: angle $= 360° \div$ order.
- Saying a figure with order 1 has rotational symmetry; order 1 means only the trivial full turn.
- Drawing the mirror image at the wrong distance from the mirror line or on the same side of it.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.